

Assignment 1: Row vs Columnar CSV Data

Prompt:

“take a look at the various honda models that are in circulation at the moment (i.e. civics, accords, odysseys, etc.). i would like for you to generate a script that itself generates 2 different csv files. the first csv file will be a standard one, with 10,000 rows, and 4 columns dedicated to either a car's VIN, which is a 10-digit alphanumeric ID, its model, the price, and the color of the car. the second csv file is essentially the first one transposed - now there will be 4 rows, one for each of the fields found in the first csv file, but each column will represent a new car (in other words, there will be 4 rows and 10,000 columns separated by commas).

for example, the first csv file would look something like this:

vin, model, price, color

a12bc34de5, Accord, 5000, Silver

f6g7h8i9j0, Odyssey, 10000, Blue . . .

while the second file would look something like this:

a12bc34de5, f6g7h8i9j0, ...

Accord, Odyssey ,...

5000, 10000, ...

Silver, Blue, ...

afterwards, i want you to generate two different python scripts: the first one will compute the time it takes to read through each row in the first csv file, extract the third element (i.e. price) of the row, and compute the average value. the second one will do the same but for the second csv file - it will only extract the third row from the second csv file, and calculate the average value of that row, in the end returning the time it took to compute that average.”

CSV Files:

First CSV File (Preview):

```
vin,model,price,color  
0wlpab10eg,CR-V Hybrid,36045,Beige  
1dy68ry236,Prologue,51406,Red  
cxql59nf47,HR-V,28703,Red  
kd46j2lvgt,Accord Hybrid,32146,White
```

Second CSV File (Preview):

```
0wlpab10eg,1dy68ry236,cxql59nf47,kd46j2lvgt,xkipx2
```

Visualizing I/O:

Standard CSV Analysis:

```
Standard CSV Analysis (Row-based):  
Total cars processed: 10000  
Average price: $36,503.00  
Time taken: 0.009502 seconds
```

Transposed CSV Analysis:

```
Transposed CSV Analysis (Column-based):  
Total cars processed: 10000  
Average price: $36,503.00  
Time taken: 0.007395 seconds
```

Silver Car Count (Standard CSV):

```
Standard CSV Analysis (Row-based) - Silver Car Count:  
Total cars processed: 10000  
Silver cars found: 961  
Percentage: 9.61%  
Time taken: 0.007832 seconds
```

Silver Car Count (Transposed CSV):

Transposed CSV Analysis (Column-based) - Silver Car Count:

Total cars processed: 10000

Silver cars found: 961

Percentage: 9.61%

Time taken: 0.005784 seconds

Analysis:

Going through the second csv file (transposed csv) was faster than going through the first csv file because there was less wasted data in traversing the second csv file - the first csv file required going through each row and discarding around 75% of the total data in that row, but for the second csv file, no data was wasted.

AI Analysis:

I used Anthropic's Claude Sonnet 4.5 model to generate my results. This LLM was very effective in generating simple python scripts that did what they had to without much bloat. I did not need to provide manual adjustments to the model output.

Critical Thinking:

- We used the three tier structure in this assignment (conceptual tier being the first line of the first CSV file which defined what the structure of the database would be, implementation tier being the actual data found within the CSV file, and hardware tier being the device, in this case my device, which was used to store the data), as well as the schema/instance concepts, with the schema being the structure of either csv file and instances being either a row or column respectively.
- If i wanted to update the price of a model, then using the first csv format would be better than the second one because to do so, you would need to find all entries with a specific model, and update their corresponding price. The first format makes it easier to access multiple types of data associated with a specific entry, so the first csv file would be preferred for this method.
- To know the average price of all honda civics, i would tell the AI to generate a method that iterates through the first csv file to find the average price of all civics and civic hybrids (the specific model output here has both hybrid and non-hybrid varieties for the civic).